

DimSplat: A Real-Time Diminished Reality System for Revisiting Environments Using Gaussian Splats in Mobile WebXR

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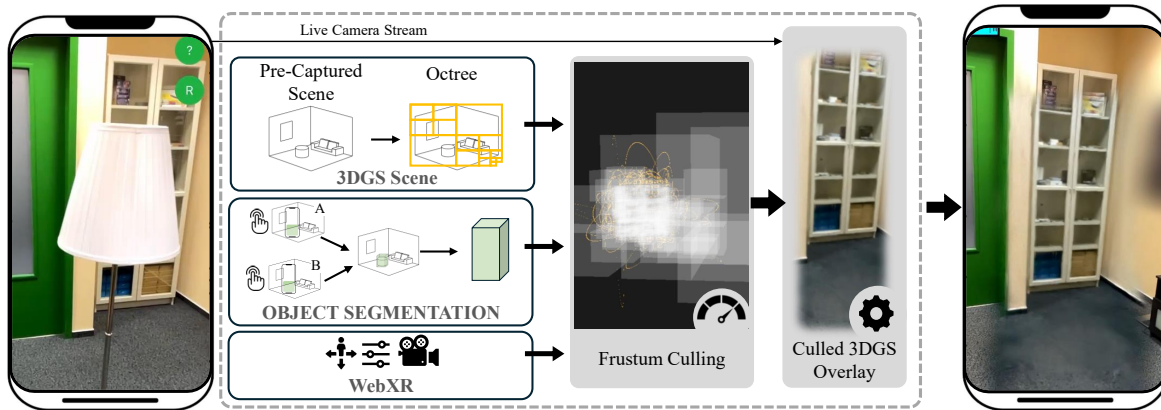


Figure 1: The overview of our **DimSplat** system to remove real objects with 3D Gaussian Splatting (3DGS) [3] in AR using WebXR. This is achieved by creating a bounding volume via two selection points from different perspectives. Together with a pre-captured 3DGS scene, reorganized in an octree structure, the system performs a fast culling to select only the necessary splats. In combination with a live video stream, the system overlays the culled 3DGS scene to diminish the selected real objects.

ABSTRACT

Diminished Reality (DR) allows for the selective removal or transformation of real-world objects from a user's field of view. In certain cases, it becomes essential to revisit environments and observe them in their original state. While Smartphone and Web-based approaches provide intuitive and convenient AR platforms, it also faces challenges like limited resources. This work introduces a DR system using Gaussian splats for real-time object removal on WebXR-enabled devices. We achieve up to 60 FPS while maintaining high visual fidelity, validated with different image quality metrics, offering a scalable solution for revisiting and manipulating temporally altered environments.

Index Terms: Diminished Reality, 3D Gaussian Splats, WebXR, Real-time Rendering, 3D Object Removal, Mobile Devices

1 INTRODUCTION

Diminished Reality (DR) is a subset of mixed reality technologies, allowing the selective removal or transformation of objects in the real world from the user's field of view. While Augmented Reality (AR) enriches environments by adding digital elements, DR simplifies or adjusts the real-world scene by reducing visual information

[5]. A special scenario for DR lies in on-site revisiting environments that have undergone temporal changes, such as preservation of heritage, renovation planning, construction, or privacy-sensitive environments [4, 7]. For instance, interior designers may require visualizations of spaces without currently existing furniture from hand-held devices for easy use. Current DR methods face challenges with dynamic lighting, where inconsistent illumination disrupts visual fidelity; occlusions, which hinder accurate object removal; and high computational demands that limit real-time performance on consumer devices [6, 2]. This work addresses these challenges by introducing a novel DR system designed to revisit environments over time. Our system uses 3D Gaussian splatting [3], a rendering approach using compact 3D primitives that enable efficient scene representation while maintaining high visual fidelity. Using pre-captured scene data, the system facilitates precise object registration, real-time object removal with fast splat culling via an octree data structure, and manipulation via optimized WebXR compatible pipelines (see Fig. 1). We evaluated performance and visual consistency with the Structural Similarity Index (SSIM) and Learning Perceptual Image Patch Similarity (LPIPS) metrics in different scenes, achieving up to 60 FPS on mobile devices in motion.

2 METHOD

For preparation, the scene must be captured in a prior state using the Gaussian splat generation by Kerbl et al. [3]. The captured data is converted into a point cloud and subsequently compressed into the *.splat* format. This lightweight representation minimizes data size while maintaining high visual fidelity, enabling smooth WebXR rendering. The **process of diminishing** objects needs spatial con-

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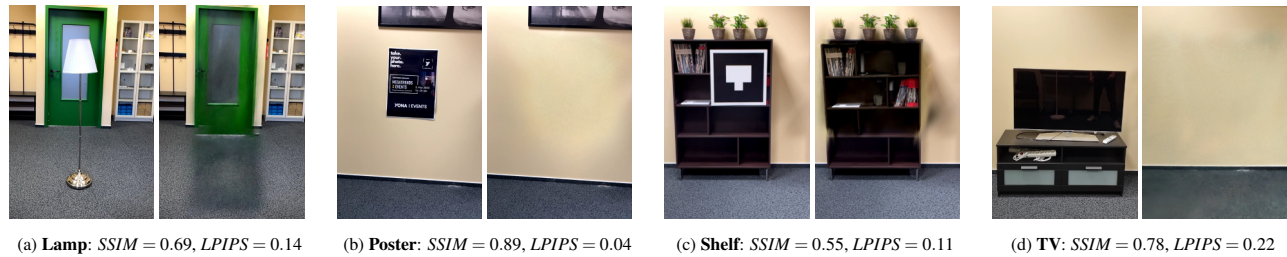


Figure 2: Comparison of various diminished objects. The image left shows the untouched scene, while The right image depicts the diminished scene. The Structural Similarity Index (SSIM) and Learned Perceptual Image Patch Similarity (LPIPS) scores below are calculated based on the diminished images to images where objects are physically removed (not shown). (Optimal values: $SSIM = 1.0$, $LPIPS = 0.0$)

sistency between virtual and real-world elements. Therefore, the scene registration involves the manual alignment step of the splat scene using a transparent overlay. Positional discrepancies are corrected by adjusting a positional delta, ensuring accurate integration of the virtual scene. Next, users define objects for removal by marking their diagonal corners from two perspectives on the screen, creating camera-aligned frusta in 3D space. The intersection of these frusta forms a precise bounding cuboid that encapsulates the target object. In order to select the overlaying splats and optimize rendering performance, the system employs frustum culling with an octree data structure. Hence, spatially close Gaussian splats are grouped into hierarchical octants, enabling efficient querying when a user's frustum is generated. Only overlapping octants are further evaluated, and a distance-checking algorithm filters splats outside the frustum. This approach eliminates redundant computations, ensuring that only relevant splats are rendered. To reduce artifacts at the edges, we added transparency adjustments for splats near the frustum edges. This softens the transitions and avoids abrupt boundaries, ensuring that the diminished regions blend more seamlessly with the surrounding scene.

3 DISCUSSION

We evaluated our method on the Samsung Galaxy S21 5G (*Exynos 2100, Octa Core*) and iPhone 12 (*A14 Bionic*) using four different diminished objects (see Fig. 2). The evaluation included static tests, with fixed identical positioning for the reference image capture, and dynamic tests, using handheld captures with slight movements and varying angles. LPIPS and SSIM scores are calculated by comparing reference images, where the real object has been removed, to the reconstructed counterparts generated by our system. The iPhone achieved better performance, averaging 43 FPS with a 99th percentile of 60 FPS, meeting the critical threshold in nearly all cases. The Galaxy S21 averaged 18 FPS with a 99th percentile of 29 FPS, indicating difficulty maintaining high frame rates during motion. In all evaluations, our **DimSplat** system uses frustum culling within the octree data structure to ensure real-time capability. Reduction rates depend on the bounding volume and camera perspective; for instance, the lamp's diminished region processed $\sim 20\times$ fewer splats, reducing from $\sim 900K$ to $\sim 45K$.

Although the system performs effectively in uniform environments, such as the *Poster* in Figure 2b and *TV* in Figure 2d ($SSIM_{poster} = 0.89$, $SSIM_{TV} = 0.78$), its true advantage lies in handling more complex backgrounds. In contrast to other inpainting methods, which often "hallucinate" details or only approximate the background, our approach leverages pre-captured scene data, providing accurate and complete background information. This allows for the restoration of elements that were removed, as demonstrated by the diminished real objects (e.g., the shelf behind the lamp in Fig. 1). In simpler and more uniform scenes, the system achieves higher quality measures due to reduced alignment errors. However,

one limitation of the system is its reliance on pre-recorded scenes, which restricts real-time adaptability to dynamic lighting and live object selection. Although Gaussian splatting improves efficiency, our method lacks fine-grained control for individual object manipulation. Additionally, blending artifacts remain a significant limitation, particularly in terms of visual quality, further emphasizing the need for refinement in these areas.

4 CONCLUSION AND FUTURE WORK

This work introduces a mobile Diminished Reality (DR) system that leverages Gaussian splats and WebXR to achieve real-time object removal while maintaining reliable visual fidelity. By integrating frustum culling with an octree data structure, the system achieves efficient rendering, significantly reducing computational demands while maintaining up to 60 FPS on consumer devices. The suggested method demonstrates its efficacy in both simple and complex settings. Quantitative metrics, such as SSIM and LPIPS, confirm the reliable perceptual quality of the system. Although effective, the system highlights several opportunities for improvement. Its reliance on pre-recorded scenes limits dynamic adaptability, and challenges in precise object selection highlight the need for further refinement. Future research could address these limitations by integrating adaptive lighting models, facilitating real-time object segmentation, and advancing Gaussian splatting techniques to allow greater flexibility in dynamic environments [1].

ACKNOWLEDGMENTS

The authors wish to thank *YONA Systems GmbH* for their supportive work on this project.

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